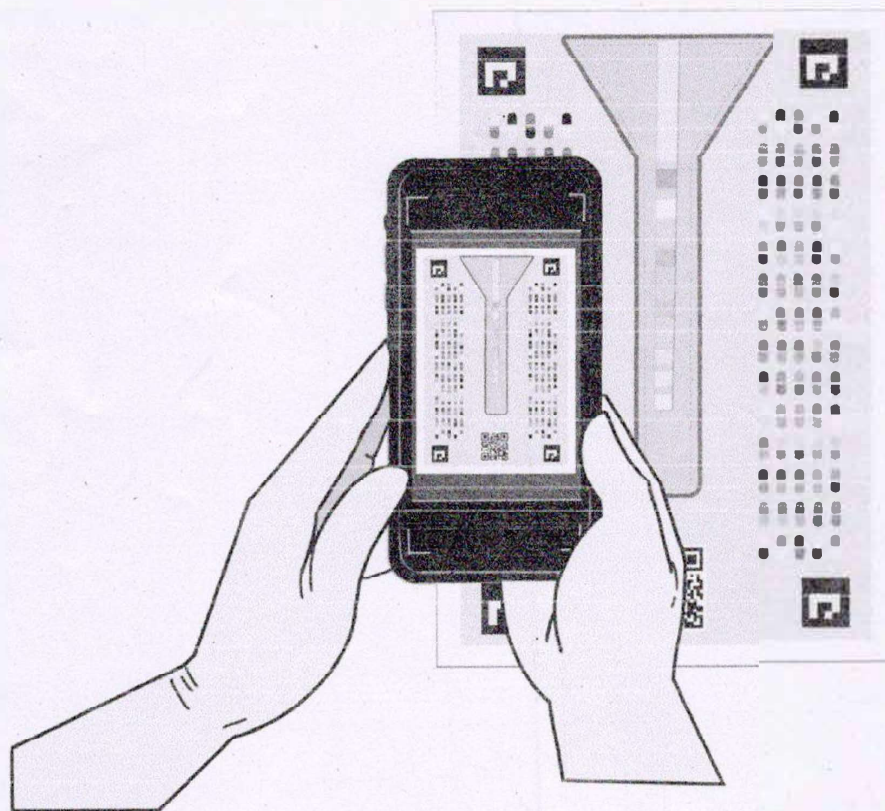


Clinical Trial cum Performance Evaluation

Nov- Dec 2025

Clinical Trial cum Performance Evaluation of the BillionTests Urine Strip Scanner Mobile Application (TRL-7) for Interpretation of Multi-Analyte Urine Dipsticks



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Protocol ID: BT-IN-P1-001

Version: 1.0

Study Completion Date: 01-12-2025

Sponsor: VisionXcel Technologies Pvt. Ltd.

SPONSOR DETAILS

VisionXcel Technologies Pvt. Ltd.

Registered Office Address: Plot No-11, KH. No.-501SA-502SA, Shukla Vihar, Devpur, Rajajipuram,
Lucknow, Uttar Pradesh-226017

Email: pranav@excl.ai

Phone: +91 9651891556

STUDY SITE:

Primary Clinical Laboratory

Name: City Imaging & Clinical Labs LLP

Accreditation: NABL

NABL Accreditation No.: MC-6723

ISO 15189:2022 Accredited

Laboratory License: AAC-8193

Location: 4B/17-18, Tilak Nagar, New Delhi-110018

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INVESTIGATOR & STUDY TEAM

Principal Investigator (PI)

Name: Dr. Sunita Kapoor

Qualification: MD / Pathology (Registered with DMC)

Affiliation: [Hospital/Lab]

Experience e.g.: ≥ 5 years in clinical diagnostics

Accreditation e.g.: DMC Registration No. 6273

Signature: [Signature]

Date: 8/Dec/2025

Clinical Supervisor

Name: Dr. Vartika Sachdeva

Qualification: MBBS/MD(Pathology)

Role: Clinical oversight, safety oversight

Accreditation e.g.: DMC Registration No.95041

Signature: [Signature]

Date: 08.12.2025

Laboratory Technician

Name: Paramjeet Solanki

Qualification: BMLT / DMLT

Experience e.g.: ≥ 3 years

Role: Conduct reference analyser testing and strip handling

Signature: [Signature]

Date: 08/Dec/2025



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BRIEF SUMMARY

VisionXcel Technologies Pvt. Ltd. (Lucknow, Uttar Pradesh) requested City Imaging & Clinical Labs LLP to conduct a clinical trial and performance evaluation of the BillionTests Urine Strip Scanner Mobile Application (TRL-7) for interpreting multi-analyte urine dipsticks.

The BillionTests system comprises a mobile application, BT Card, standard third-party urine dipstick strips, and an AI cloud server. The smartphone camera captures images of urine samples on the BT Card, which are processed locally and then sent to the cloud. Results appear on-screen as PDFs that can be shared via WhatsApp/email or printed, the average time between capturing an image and displaying the results is 10-15 seconds, all data securely stored using encryption and blockchain.

City Imaging & Clinical Labs LLP evaluated the device starting November 2025 using a diverse sample set (negative/positive results, various ages, both genders). Testing was conducted across multiple phone devices, lighting conditions, and angles to assess accuracy. Performance was benchmarked against the Roche Cobas U411 Semi automated Urine Analyzer and manual analysis.

The evaluation covered 10 urinalysis parameters: Glucose, Protein, Ketones, Blood, pH, Leukocytes, Nitrite, Urobilinogen, Specific Gravity, and Bilirubin.

Validation Results-

1. BillionTests Urine Strip Analyzer validation against Roche Cobas U411 Analyzer and Roche Combur-Test strip

Overall Concordance: **98.52%**

2. BillionTests Urine Strip Analyzer validation against multi-brand urine test strips compatibility using manual analysis

Overall Concordance: **99.8%**

3. BillionTests Urine Strip Analyzer validation against multi-brand urine test strips' time effect study

Best Time Points: **1, 4, 5 minutes (98% concordance)**

Minimum Concordance: **96% (at 2-3 minutes)**

Concordance Range: **96% - 98%**